

14 - Hydrocarbons

28/12/24

Alkanes ★ useful as fuels ★

- saturated hydrocarbons with the general formula $[C_nH_{2n+2}]$
- insoluble in water, and boiling point increases with size of molecule
- branched alkanes are more volatile & have a lower BP than its isomer that is a straight-chained alkane

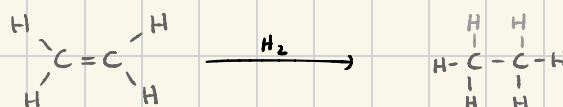
unreactivity of alkanes

- alkanes are generally unreactive due to lots of energy needed to break strong C-C & C-H bonds
- alkanes are non-polar because carbon & hydrogen have nearly the same electronegativity
- this means that they can only dissolve in non-polar solvents, & cannot react with polar reagents like nucleophiles & electrophiles

Production of alkanes

i) hydrogenation to alkenes

- addition of hydrogen to an alkene
- reagents & conditions: $H_2(g)$, Pt/Ni catalyst, & heat



ii) cracking longer chain alkanes (can also form alkenes)

- used to obtain more useful alkanes (and alkenes) of lower M_r from heavier crude oil fractions
- reagents & conditions: heat long-chain alkane with Al_2O_3

Combustion of alkanes

i) complete combustion

alkane + ^{excess} oxygen → carbon dioxide + water

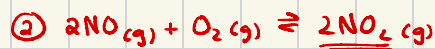
ii) incomplete combustion

alkane + ^{limited} oxygen → carbon monoxide + water

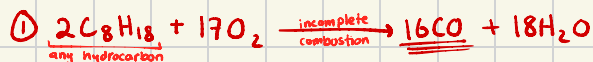
toxic gas

⇒ carbon monoxide is formed via incomplete combustion in car engines as well

releases acidic nitrogen oxides (NO & NO_2) which contribute to acid rain



releases carbon monoxide which binds to haemoglobin in RBC's & removes its ability to bind with oxygen, so your body doesn't get enough oxygen



carcinogens (cause cancer)

releases unburnt hydrocarbons (volatile organic compounds) from combustion of alkanes which can form PAN, a contributor to photochemical smog

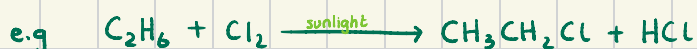
car engine

⇒ to reduce the production of NO , NO_2 & CO , catalytic converters are fitted in cars & they:

- reduce nitrogen oxides to form harmless nitrogen gas
- oxidise carbon monoxide to form carbon dioxide ← non-toxic but still a pollutant which causes global warming
- oxidise unburnt hydrocarbons to form carbon dioxide & water

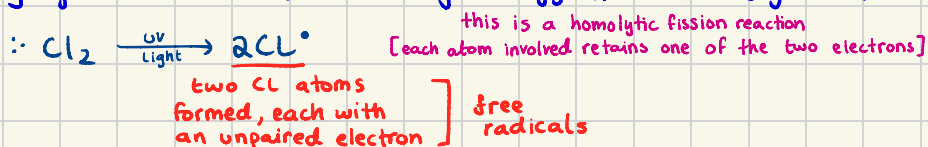
Free-radical substitution

- alkanes undergo free-radical substitution with halogens (Cl_2 or Br_2) in sunlight (presence of UV light)
- a halogen free-radical replaces a hydrogen of an alkane



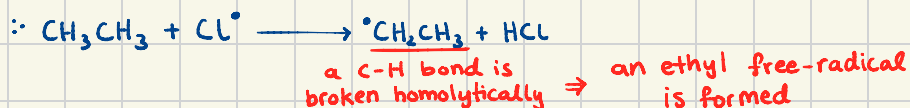
① initiation step

⇒ breaking of the $\text{Cl}-\text{Cl}$ bond using energy from UV light in sunlight

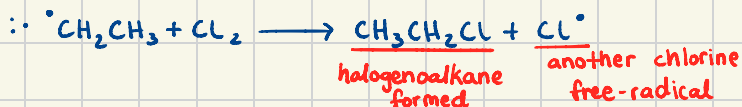


② propagation step

⇒ free radicals are very reactive; they attack the normally unreactive alkanes



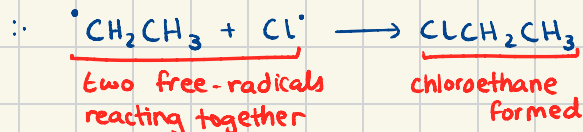
⇒ the ethyl free-radical can then attack a chlorine molecule - forming chloroethane & a chlorine free-radical



⇒ the chlorine free-radical attacks another ethane molecule; an ethyl free-radical is formed which attacks another chlorine atom, regenerating a chlorine free-radical and so on.

③ termination step

⇒ when two free-radicals meet, they will react to form a single molecule. No free-radicals are made in that case to carry on the reaction sequence, and as a result, the reaction stops



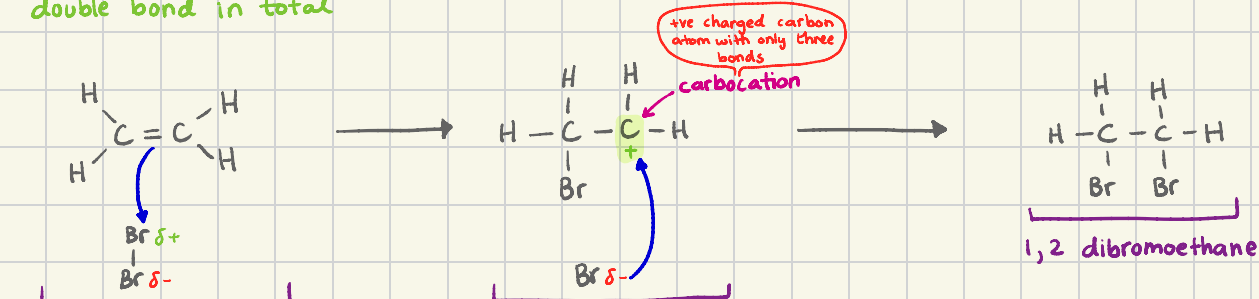
Alkenes

- unsaturated hydrocarbons with at least one C=C bond & the general formula $[C_nH_{2n}]$
- physical properties are similar to alkanes
- can exhibit both structural & stereoisomerism (due to rotation-restricting double bond)

Test for alkenes

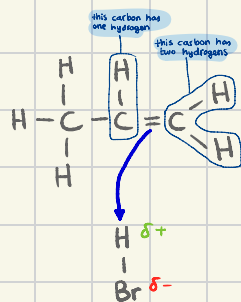
- the presence of C=C bonds can be tested by aqueous bromine
- if an unsaturated compound is put in bromine water, color changes from orange-brown $\rightarrow$ colorless

high electron density $C \equiv C$ σ bond & a π bond
 there are four electrons in this double bond in total $\leftarrow$ this makes an alkene open to attack by electrophiles



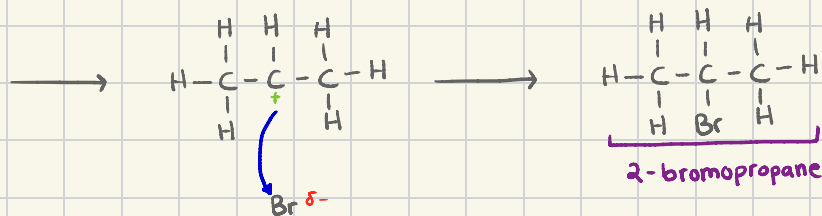
when the Br_2 molecule is near ethene, the electrons in the nearer Br atom repel the double bond. This makes the nearer Br atom slightly +ve & the further Br atom slightly -ve.

the nearer Br atom bonds with one of the carbons, & the other slightly negative Br atom is attracted to the other carbon which is now missing an electron, due to elimination of double bond, & is thus positive



Markovnikov's rule

in an addition reaction of hydrogen halide & an alkene, the hydrogen goes to the carbon that already has more hydrogen atoms



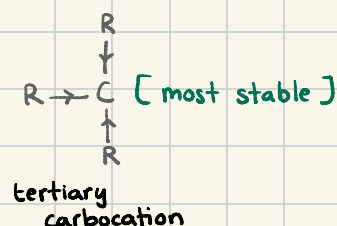
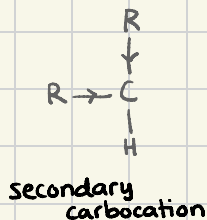
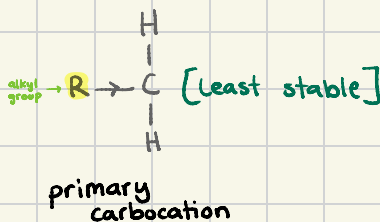
Carbocations

$\rightarrow$ three types: primary, secondary & tertiary

alkyl groups attached to carbocations are known as 'electron-donating groups'

alkyl groups push e^- away from themselves to positively charged carbon atoms

this causes the carbocation to become less +ve. Once the charge spreads around the carbon, it becomes more stable



Production of alkenes

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i) dehalogenation / elimination

- removal of a H & a halogen atom from halogenoalkanes, can form alkenes
- reagents & conditions: ethanolic NaOH with heat

ii) dehydration of alcohols

- removal of a H & OH from adjacent carbons, produces H₂O and a C=C bond can be formed
- reagents & conditions: heated catalyst (Al₂O₃) or conc. acid (conc. H₂SO₄)

iii) cracking longer chain alkanes

- used to obtain more useful alkenes (and alkanes) of lower M_r from heavier crude oil fractions
- reagents & conditions: heat long-chain alkane with Al₂O₃

Addition reaction of alkenes] electrophilic addition

i) addition of hydrogen

- reagents & conditions: Pt/Ni catalyst and heat



ii) addition of steam

- reagents & conditions: H₂O(g), H₃PO₄ catalyst & high temp./pressure



iii) addition of hydrogen halides

- reagents & conditions: conc. solution of HX & at room temperature



iv) addition of halogens

- reagents & conditions: room temperature

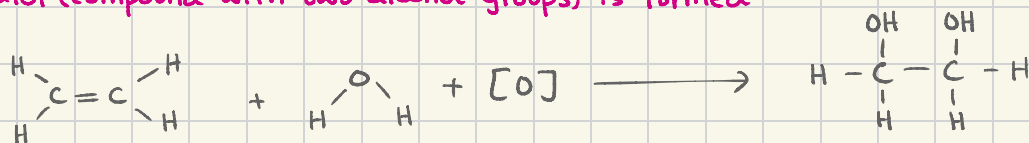


Oxidation reactions of alkenes

→ alkenes can be oxidised by acidified potassium manganate (VII) [KMnO₄(aq)]

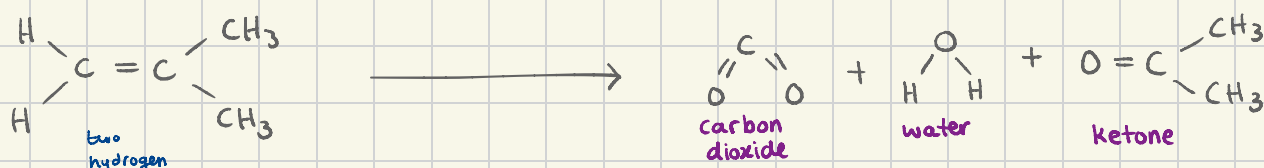
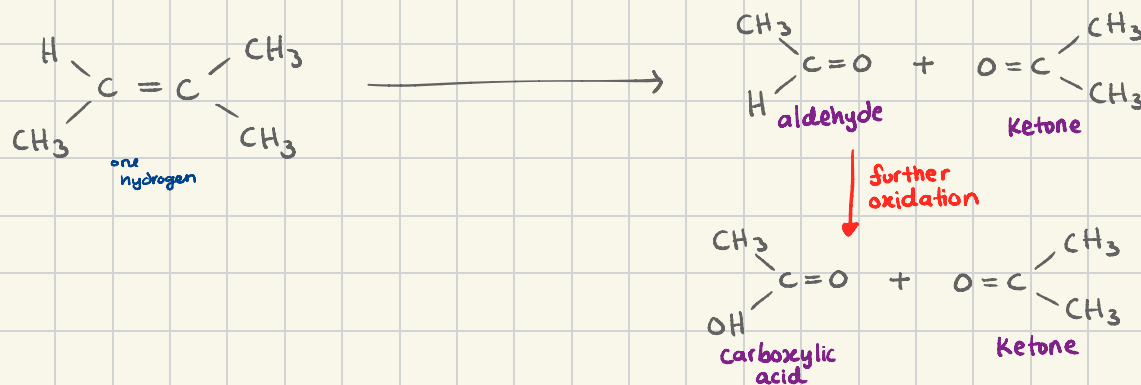
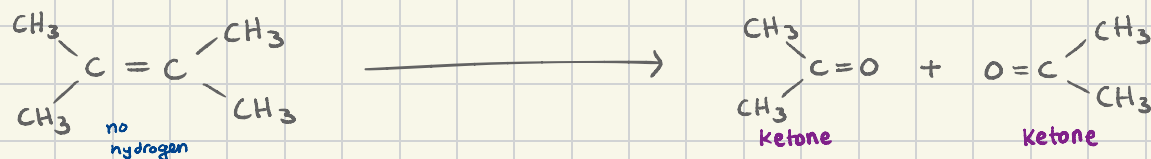
* in cold dilute acidified KMnO₄

- a diol (compound with two alcohol groups) is formed



* in hot concentrated acidified KMnO_4

- in these harsh conditions, the $\text{C}=\text{C}$ double bond of the alkene is broken completely
- the initial $\text{O}-\text{H}$ groups in the diol, further oxidise to ketones, aldehydes, carboxylic acids or carbon dioxide



SUMMARY

Starting material	Type of product
No hydrogens on either carbon	ketone
1 hydrogen on either carbon	aldehyde
↳ further oxidation of this	carboxylic acid
2 hydrogens on either carbon	carbon dioxide + water

Addition polymerisation

- a polymer is a long-chain molecule made up of many repeating units of small, reactive molecules called monomers

